The symmetric Group

Shiyu Dong

November 7, 2024

outline

1. Permutation
2. Partitions And conjugate classes
3. Young Tableaux
4. Antisymmetrizer

Permutation

$$P = \begin{pmatrix} 1 & 2 & \cdots & n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix} \tag{1}$$

$$Q = \begin{pmatrix} 1 & 2 & \cdots & n \\ q_1 & q_2 & \cdots & q_n \end{pmatrix} \tag{2}$$

$$PQ = \begin{pmatrix} 1 & 2 & \cdots & n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix} \begin{pmatrix} 1 & 2 & \cdots & n \\ q_1 & q_2 & \cdots & q_n \end{pmatrix} \tag{3}$$

$$= \begin{pmatrix} q_1 & q_2 & \cdots & q_n \\ r_1 & r_2 & \cdots & r_n \end{pmatrix} \begin{pmatrix} 1 & 2 & \cdots & n \\ q_1 & q_2 & \cdots & q_n \end{pmatrix} \tag{4}$$

$$= \begin{pmatrix} 1 & 2 & \cdots & n \\ r_1 & r_2 & \cdots & r_n \end{pmatrix} \tag{5}$$

cycle

m-cycle

$$\begin{pmatrix} 1 & 2 & \cdots & m-1 & m \\ 2 & 3 & \cdots & m & 1 \end{pmatrix} \equiv \begin{pmatrix} 1 & 2 & \cdots & m-1 & m \end{pmatrix} \tag{6}$$

2-cycle is called *transpositon*

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \quad \cdots \quad \begin{pmatrix} 1 & i \end{pmatrix} \quad \begin{pmatrix} i & j \end{pmatrix} \tag{7}$$

Parity

$$\begin{aligned}\Delta(x_1, x_2, \cdots, x_n) = (x_1 - x_2)(x_1 - x_3)\cdots(x_1 - x_n) \\ \times(x_2 - x_3)\cdots(x_2 - x_n)\cdots \\ \times(x_{n-1} - x_n)\end{aligned} \tag{8}$$

$$P\Delta = (-1)^{\sigma_p}\Delta \tag{9}$$

Decomposition

Every permutation can be expressed as a pdoduct of cycles
Example

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 5 & 1 & 3 & 6 \end{pmatrix} \equiv \begin{pmatrix} 1 & 2 & 4 \end{pmatrix} \begin{pmatrix} 3 & 5 \end{pmatrix} \begin{pmatrix} 6 \end{pmatrix} \tag{10}$$

Every permutation can be expressed as a product of transposition
A m-cycle can be decomposed into product of m-1 transposition

Decomposition

$$\begin{pmatrix} 1 & 2 & 3 & \cdots & m-1 & m \\ 1 & 2 & 3 & \cdots & m-1 & m \end{pmatrix} \tag{11}$$

$$(1m)\begin{pmatrix} 1 & 2 & \cdots & m-1 & m \\ 1 & 2 & \cdots & m-1 & m \end{pmatrix} = \begin{pmatrix} 1 & 2 & \cdots & m-1 & m \\ m & 2 & \cdots & m-1 & 1 \end{pmatrix} \tag{12}$$

$$(1m{-}1)(1m)\begin{pmatrix} 1 & 2 & \cdots & m-1 & m \\ 1 & 2 & \cdots & m-1 & m \end{pmatrix} = \begin{pmatrix} 1 & 2 & \cdots & m-1 \\ m-1 & 2 & \cdots & m \end{pmatrix} \tag{13}$$

Partition and conjugate class

$$Q = TPT^{-1} \tag{14}$$

Theorem

Two permutation in S_n are conjugate if and only if they have the same cycle structure(they have the same number of cycles of equal length)

Proof

Conjugate $\implies$ same cycle structure

$$Q = TPT^{-1} = \begin{pmatrix} 1 & 2 \cdots & m \\ t_1 & t_2 \cdots & t_m \end{pmatrix} \begin{pmatrix} 1 & 2 \cdots & m \\ p_1 & p_2 \cdots & p_m \end{pmatrix} \begin{pmatrix} t_1 & t_2 \cdots \\ 1 & 2 \cdots \end{pmatrix} \tag{15}$$

$$= \begin{pmatrix} t_1 & t_2 & \cdots & t_m \\ p_{t_1} & p_{t_2} \cdots & p_{t_m} & \end{pmatrix} \tag{16}$$

Relabeling the index.

Cycle structure

$$(\nu) = (1^{\nu_1} 2^{\nu_2} \cdots m^{\nu_m}) \tag{17}$$

$$\nu_1 + 2\nu_2 + \cdots + m\nu_m = m \tag{18}$$

$$\lambda_1 = \nu_1 + \nu_2 + \cdots + \nu_m \tag{19}$$

$$\lambda_k = \nu_k + \nu_2 + \cdots + \nu_m \tag{20}$$

$$\lambda_1 + \lambda_2 + \cdots \lambda_m = m \tag{21}$$

$$\nu_k = \lambda_k - \lambda_{k+1} \tag{22}$$

partition of integer n:

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = n \quad \lambda_1 \geq \lambda_2 \cdots \lambda_n$$

Symmetry and Antisymmety

In this article, we explore a connection between representations of the symmetric group S_n and combinatorial objects called Young tableaux. We define Young tableaux in Section 4.2, but for now, it suffices to say that they are fillings of a certain configuration of boxes with entries from $\{1, 2, \ldots, n\}$, an example of which is shown below.

$$\begin{array}{|c|c|c|}\hline 1 & 2 & 4 \\ \hline 3 & 5 & 6 \\ \hline 7 & 8 \\ \cline{1-2} 9 \\ \cline{1-1}\end{array}$$

So how are representations of S_n related to Young tableau? It turns out that there is a very elegant description of irreducible representations of S_n through Young tableaux. Let us have a glimpse of the results. Recall that there are three irreducible representations of S_3. It turns out that they can be described using the set of Young diagrams with three boxes. The correspondence is illustrated below.

trivial representation sign representation standard representation

It is true in general that the irreducible representations of S_n can be described using Young diagrams of n boxes! Furthermore, we can describe a basis of each irreducible representation using standard Young tableaux, which are numberings of the boxes of a Young diagram with $1, 2, \ldots, n$ such that the rows and columns are all increasing. For instance, the bases of the standard representation of S_3 correspond to the following two standard Young tableaux:

$$\begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1}\end{array} \qquad \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1}\end{array}$$

Figure:

$$\hat{\mathcal{A}}^{(n)}(1,2,\cdots,n) = \frac{1}{n!}\sum_{P}(-1)^{\sigma_p}\hat{P} \tag{23}$$

$$\hat{Q}\hat{\mathcal{A}}^{(n)} = (-1)^{\sigma_Q}\hat{\mathcal{A}}^{(n)} \tag{24}$$

$$\hat{\mathcal{A}}^{(n)}\hat{\mathcal{A}}^{(n)} = \hat{\mathcal{A}}^{(n)} \tag{25}$$

Proof

$$\hat{Q}\hat{\mathcal{A}}^{(n)} = \frac{1}{n!}\sum_{P}(-1)^{\sigma_p}QP \tag{26}$$

$$= \frac{1}{n!}\sum_{p}(-1)^{\sigma_Q}(-1)^{\sigma_{QP}}QP \tag{27}$$

$$= (-1)^{\sigma_Q}\frac{1}{n!}\sum_{qp}(-1)^{\sigma_{QP}}QP \tag{28}$$

$$= (-1)^{\sigma_Q}\hat{\mathcal{A}}^{(n)} \tag{29}$$

$$\hat{\mathcal{A}}^{(n)}\hat{\mathcal{A}}^{(n)} = (\frac{1}{n!}\sum_{Q}(-1)^{\sigma_Q}Q)(\frac{1}{n!}\sum_{P}(-1)^{\sigma_P}P) \tag{30}$$

$$= \frac{1}{n!}\sum_{Q}(-1)^{\sigma_Q}(-1)^{\sigma_Q}\hat{\mathcal{A}}^{(n)} \tag{31}$$

$$= \hat{A}^{(n)} \tag{32}$$